

Static latch-up

The following complementary static tests are required on three parts to assess the latch-up performance:

- A supply overvoltage is applied to each power supply pin.
- A current injection is applied to each input, output, and configurable I/O pin.

These tests are compliant with EIA/JESD 78E IC latch-up standard.

Table 46. Electrical sensitivities

Symbol	Parameter	Conditions	Class
LU	Static latch-up class	$T_A = 130^\circ\text{C}$ conforming to JESD78	Level II A

5.3.13 I/O current injection characteristics

As a general rule, the current injection to the I/O pins, due to external voltage below V_{SS} or above V_{DDIOx} (for standard, 3.3 V-capable I/O pins) must be avoided during normal product operation. However, in order to give an indication of the robustness of the microcontroller if abnormal injection accidentally happens, some susceptibility tests are performed on a sample basis during the device characterization.

Functional susceptibility to I/O current injection

While a simple application is executed on the device, the device is stressed by injecting current into the I/O pins programmed in floating-input mode. While this current is injected into the I/O pin, one at a time, the device is checked for functional failures.

The failure is indicated by an out-of-range parameter, such as an ADC error above a certain limit (higher than 5 LSB TUE), out of conventional limits of induced leakage current on adjacent pins (out of the $5\ \mu\text{A}/+0\ \mu\text{A}$ range), or other functional failure (for example reset occurrence or oscillator frequency deviation).

The characterization results are given in the table below. The negative induced leakage current is caused by the negative injection. The positive induced leakage current is caused by the positive injection.

Table 47. I/O current injection susceptibility

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.
 Evaluated by characterization - Not tested in production.

Symbol	Description	Functional susceptibility		Unit
		Negative injection	Positive injection	
I_{INJ}	Injected current on PA4 pins	0	0	mA
	Injected current on PB13, PB14, PB15, PD8, PD9, PD10, PD11, PD12, PD13, PE0, and PE1 pins	0	N/A	
	Injected current on all other pins	5	N/A	